HW4(optional) B063012054 林祐安

```python
def initialize_parameters(n_x, n_h, n_y):
    """
    Argument:
    n_x -- size of the input layer
    n_h -- size of the hidden layer
    n_y -- size of the output layer

    Returns:
    parameters -- python dictionary containing your parameters:
                    W1 -- weight matrix of shape (n_h, n_x)
                    b1 -- bias vector of shape (n_h, 1)
                    W2 -- weight matrix of shape (n_y, n_h)
                    b2 -- bias vector of shape (n_y, 1)
    """

    np.random.seed(1)

    ### START CODE HERE ### (≈ 4 lines of code)
    W1 = np.random.randn(n_h,n_x)*0.01
    b1 = np.zeros([n_h,1])
    W2 = np.random.randn(n_y,n_h)*0.01
    b2 = np.zeros([n_y,1])
    ### END CODE HERE ###

    assert(W1.shape == (n_h, n_x))
    assert(b1.shape == (n_h, 1))
    assert(W2.shape == (n_y, n_h))
    assert(b2.shape == (n_y, 1))

    parameters = {"W1": W1,
                  "b1": b1,
                  "W2": W2,
                  "b2": b2}

    return parameters

```

結果：

```
W1 = [[ 0.01624345 -0.00611756]
 [-0.00528172 -0.01072969]]
b1 = [[0.]
 [0.]]
W2 = [[ 0.00865408 -0.02301539]]
b2 = [[0.]]
```

建立初始值，np.random.randn()的隨機值取於均數=0，標準差=1 的常態分佈，為了配合梯度更新，需將值*0.01。

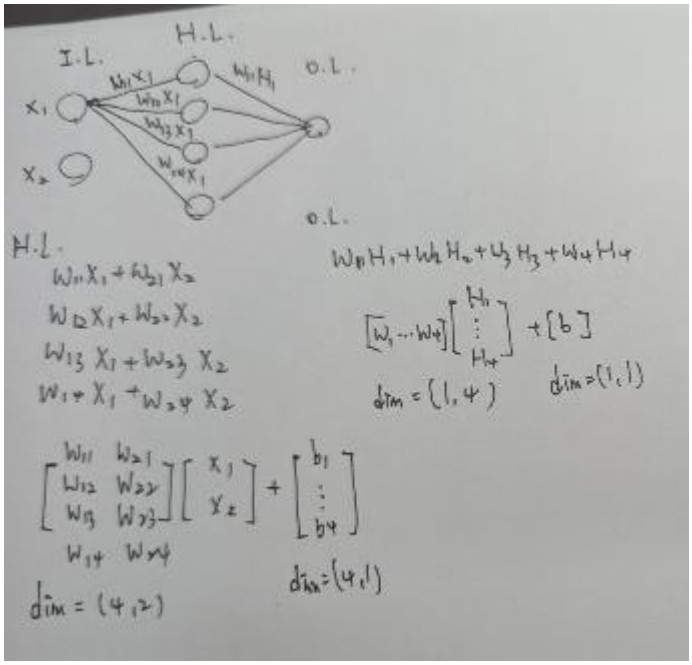
```python
def initialize_parameters_deep(layer_dims):
    """
    Arguments:
    layer_dims -- python array (list) containing the dimensions of each layer in our network

    Returns:
    parameters -- python dictionary containing your parameters "W1", "b1", ..., "WL", "bL":
                    Wl -- weight matrix of shape (layer_dims[l], layer_dims[l-1])
                    bl -- bias vector of shape (layer_dims[l], 1)
    """

    np.random.seed(3)
    parameters = {}
    L = len(layer_dims)            # number of layers in the network

    for l in range(1, L):
        ### START CODE HERE ### (≈ 2 lines of code)
        parameters['W' + str(l)] = np.random.randn(layer_dims[l], layer_dims[l-1])*0.01
        parameters['b' + str(l)] = np.zeros([layer_dims[l], 1])
        ### END CODE HERE ###

        assert(parameters['W' + str(l)].shape == (layer_dims[l], layer_dims[l-1]))
        assert(parameters['b' + str(l)].shape == (layer_dims[l], 1))


    return parameters
```

結果：

```
W1 = [[ 0.01788628  0.0043651   0.00096497 -0.01863493 -0.00277388]
 [-0.00354759 -0.00082741 -0.00627001 -0.00043818 -0.00477218]
 [-0.01313865  0.00884622  0.00881318  0.01709573  0.00050034]
 [-0.00404677 -0.0054536  -0.01546477  0.00982367 -0.01101068]]
b1 = [[0.]
 [0.]
 [0.]
 [0.]]
W2 = [[-0.01185047 -0.0020565   0.01486148  0.00236716]
 [-0.01023785 -0.00712993  0.00625245 -0.00160513]
 [-0.00768836 -0.00230031  0.00745056  0.01976111]]
b2 = [[0.]
 [0.]
 [0.]]
```

討論：

一開始對於初始化參數中的維度，沒有頭緒，為何要迴圈？為什麼裡面的維度是 layer_dim[l]，動手畫了圖之後較好想像。



以一個 layer_dims=(2,4,1)為例，將每個層的參數用矩陣表示並整理，就能歸納出一個規則，W 權重的維度=(該層參數量,上一層參數量)，b 的維度=(該層參數量,1)。

```python
def linear_forward(A, W, b):
    """
    Implement the linear part of a layer's forward propagation.

    Arguments:
    A -- activations from previous layer (or input data): (size
    W -- weights matrix: numpy array of shape (size of current
    b -- bias vector, numpy array of shape (size of the current

    Returns:
    Z -- the input of the activation function, also called pre-
    cache -- a python dictionary containing "A", "W" and "b" ;
    """

    ### START CODE HERE ### (≈ 1 line of code)
    Z = W.dot(A)+b
    ### END CODE HERE ###

    assert(Z.shape == (W.shape[0], A.shape[1]))
    cache = (A, W, b)

    return Z, cache
```

結果：

```
Z = [[ 3.26295337 -1.23429987]]
```

```python
def linear_activation_forward(A_prev, W, b, activation):
    """
    Implement the forward propagation for the LINEAR->ACTIVATION la

    Arguments:
    A_prev -- activations from previous layer (or input data): (siz
    W -- weights matrix: numpy array of shape (size of current laye
    b -- bias vector, numpy array of shape (size of the current lay
    activation -- the activation to be used in this layer, stored a

    Returns:
    A -- the output of the activation function, also called the pos
    cache -- a python dictionary containing "linear_cache" and "act
             stored for computing the backward pass efficiently
    """

    if activation == "sigmoid":
        # Inputs: "A_prev, W, b". Outputs: "A, activation_cache".
        ### START CODE HERE ### (≈ 2 lines of code)
        Z, linear_cache = linear_forward(A_prev, W, b)
        A, activation_cache = sigmoid(Z)
        ### END CODE HERE ###

    elif activation == "relu":
        # Inputs: "A_prev, W, b". Outputs: "A, activation_cache".
        ### START CODE HERE ### (≈ 2 lines of code)
        Z, linear_cache = linear_forward(A_prev, W, b)
        A, activation_cache = relu(Z)
        ### END CODE HERE ###

    assert (A.shape == (W.shape[0], A_prev.shape[1]))
    cache = (linear_cache, activation_cache)

    return A, cache
```

結果：

```
With sigmoid: A = [[0.96890023 0.11013289]]
With ReLU: A = [[3.43896131 0.        ]]
```

提取 cache 是為了之後 backward 計算方便，每一個 forward 做一次運算就將結果值存入。
這個 function 是用來選取激活函數是 sigmoid 或 relu。

```python
def L_model_forward(X, parameters):
    """
    Implement forward propagation for the [LINEAR->RELU]*(L-1)->LINEAR->SIGMOID computation

    Arguments:
    X -- data, numpy array of shape (input size, number of examples)
    parameters -- output of initialize_parameters_deep()

    Returns:
    AL -- last post-activation value
    caches -- list of caches containing:
                every cache of linear_relu_forward() (there are L-1 of them, indexed from 0 to L-2)
                the cache of linear_sigmoid_forward() (there is one, indexed L-1)
    """

    caches = []
    A = X
    L = len(parameters) // 2                  # number of layers in the neural network

    # Implement [LINEAR -> RELU]*(L-1). Add "cache" to the "caches" list.
    for l in range(1, L):
        A_prev = A

        ### START CODE HERE ### (≈ 2 lines of code)
        A, cache = linear_activation_forward(A_prev, parameters["W"+str(l)], parameters["b"+str(l)],"relu")
        caches.append(cache)
        ### END CODE HERE ###

    # Implement LINEAR -> SIGMOID. Add "cache" to the "caches" list.
    ### START CODE HERE ### (≈ 2 lines of code)
    AL, cache = linear_activation_forward(A, parameters["W"+str(L)], parameters["b"+str(L)],"sigmoid")
    caches.append(cache)
    ### END CODE HERE ###

    assert(AL.shape == (1,X.shape[1]))

    return AL, caches
```

結果：

```
AL = [[0.17007265 0.2524272 ]]
Length of caches list = 2
```

中間的迴圈對應到(Linear->Relu)*(L-1)，使用 relu 當作激活函數，並將每次更新的值存入 cache，最後一層為 sigmoid，因此獨立於回圈之外。

```python
def compute_cost(AL, Y):
    """
    Implement the cost function defined by equation (7).

    Arguments:
    AL -- probability vector corresponding to your label predictions,
    Y -- true "label" vector (for example: containing 0 if non-cat, 1

    Returns:
    cost -- cross-entropy cost
    """

    m = Y.shape[1]

    # Compute loss from aL and y.
    ### START CODE HERE ### (≈ 1 lines of code)
    cost = -(np.sum(Y*np.log(AL)+(1-Y)*np.log(1-AL)))/m
    ### END CODE HERE ###

    cost = np.squeeze(cost)      # To make sure your cost's shape is
    assert(cost.shape == ())

    return cost

```

結果：

```
cost = 0.414931599615397
```

```python
def linear_backward(dZ, cache):
    """
    Implement the linear portion of backward propagation for a s

    Arguments:
    dZ -- Gradient of the cost with respect to the linear output
    cache -- tuple of values (A_prev, W, b) coming from the forw

    Returns:
    dA_prev -- Gradient of the cost with respect to the activati
    dW -- Gradient of the cost with respect to W (current layer
    db -- Gradient of the cost with respect to b (current layer
    """
    A_prev, W, b = cache
    m = A_prev.shape[1]

    ### START CODE HERE ### (≈ 3 lines of code)
    dW = dZ.dot(A_prev.T)/m
    db = np.sum(dZ, axis = 1, keepdims = True)/m
    dA_prev = (W.T).dot(dZ)
    ### END CODE HERE ###

    assert (dA_prev.shape == A_prev.shape)
    assert (dW.shape == W.shape)
    assert (db.shape == b.shape)

    return dA_prev, dW, db
```

結果：

```
dA_prev = [[ 0.51822968 -0.19517421]
 [-0.40506361  0.15255393]
 [ 2.37496825 -0.89445391]]
dW = [[-0.10076895  1.40685096  1.64992505]]
db = [[0.50629448]]
```

特別需要注意維度的問題， 例如 np.sum()，若沒有加入 axis=1，keepdims=True，其默認將矩陣裡的數相加，這會造成輸出維度錯誤。

```python
def linear_activation_backward(dA, cache, activation):
    """
    Implement the backward propagation for the LINEAR->ACTIVATION layer.

    Arguments:
    dA -- post-activation gradient for current layer l
    cache -- tuple of values (linear_cache, activation_cache) we store f
    activation -- the activation to be used in this layer, stored as a t

    Returns:
    dA_prev -- Gradient of the cost with respect to the activation (of t
    dW -- Gradient of the cost with respect to W (current layer l), same
    db -- Gradient of the cost with respect to b (current layer l), same
    """
    linear_cache, activation_cache = cache

    if activation == "relu":
        ### START CODE HERE ### (≈ 2 lines of code)
        dZ = relu_backward(dA,activation_cache)
        dA_prev, dW, db = linear_backward(dZ, linear_cache)
        ### END CODE HERE ###

    elif activation == "sigmoid":
        ### START CODE HERE ### (≈ 2 lines of code)
        dZ = sigmoid_backward(dA,activation_cache)
        dA_prev, dW, db = linear_backward(dZ, linear_cache)
        ### END CODE HERE ###

    return dA_prev, dW, db
```

結果：

```
sigmoid:
dA_prev = [[ 0.11017994  0.01105339]
 [ 0.09466817  0.00949723]
 [-0.05743092 -0.00576154]]
dW = [[ 0.10266786  0.09778551 -0.01968084]]
db = [[-0.05729622]]

relu:
dA_prev = [[ 0.44090989 -0.        ]
 [ 0.37883606 -0.        ]
 [-0.2298228   0.        ]]
dW = [[ 0.44513824  0.37371418 -0.10478989]]
db = [[-0.20837892]]
```

順序是(Linear->Relu)*(L-1)->Linear->Sigmoid，因此 backward 時 cache 會先提取通過激活函數的 cache，再提取 Linear 層的 cache。

```
ef L_model_backward(AL, Y, caches):
    """
    Implement the backward propagation for the [LINEAR->RELU] * (L-1) -> LINEAR -> SIGMOID group

    Arguments:
    AL -- probability vector, output of the forward propagation (L_model_forward())
    Y -- true "label" vector (containing 0 if non-cat, 1 if cat)
    caches -- list of caches containing:
                every cache of linear_activation_forward() with "relu" (it's caches[l], for l in range(L-1) i.e l = 0...L-2)
                the cache of linear_activation_forward() with "sigmoid" (it's caches[L-1])

    Returns:
    grads -- A dictionary with the gradients
             grads["dA" + str(l)] = ...
             grads["dW" + str(l)] = ...
             grads["db" + str(l)] = ...
    """
    grads = {}
    L = len(caches) # the number of layers
    m = AL.shape[1]
    Y = Y.reshape(AL.shape) # after this line, Y is the same shape as AL

    # Initializing the backpropagation
    ### START CODE HERE ### (1 line of code)
    dAL = - (np.divide(Y, AL) - np.divide(1 - Y, 1 - AL))
    ### END CODE HERE ###

    # Lth layer (SIGMOID -> LINEAR) gradients. Inputs: "AL, Y, caches". Outputs: "grads["dAL"], grads["dWL"], grads["dbL"]
    ### START CODE HERE ### (approx. 2 lines)
    current_cache = caches[L-1]
    grads["dA" + str(L)], grads["dW" + str(L)], grads["db" + str(L)] = linear_activation_backward(dAL, current_cache, "sigmoid
    ### END CODE HERE ###

    for l in reversed(range(L - 1)):
        # lth layer: (RELU -> LINEAR) gradients.
        # Inputs: "grads["dA" + str(l + 2)], caches". Outputs: "grads["dA" + str(l + 1)] , grads["dW" + str(l + 1)] , grads["d
        ### START CODE HERE ### (approx. 5 lines)
        current_cache = caches[l]
        dA_prev_temp, dW_temp, db_temp = linear_activation_backward(grads["dA" + str(l + 2)], current_cache,"relu")
        grads["dA" + str(l + 1)] = dA_prev_temp
        grads["dW" + str(l + 1)] = dW_temp
        grads["db" + str(l + 1)] = db_temp
        ### END CODE HERE ###

    return grads
```

結果：

```
dW1 = [[0.41010002 0.07807203 0.13798444 0.10502167]
 [0.         0.         0.         0.        ]
 [0.05283652 0.01005865 0.01777766 0.0135308 ]]
db1 = [[-0.22007063]
 [ 0.        ]
 [-0.02835349]]
dA1 = [[ 0.          0.52257901]
 [ 0.         -0.3269206 ]
 [ 0.         -0.32070404]
 [ 0.         -0.74079187]]
```

forward 的順序是(Linear->Relu)*(L-1)->Linear->Sigmoid，因此 backward 時激活函數 sigmoid 只需一次，剩下的層皆是 relu。

```python
def update_parameters(parameters, grads, learning_rate):
    """
    Update parameters using gradient descent

    Arguments:
    parameters -- python dictionary containing your parameters
    grads -- python dictionary containing your gradients, output of L_model_backward

    Returns:
    parameters -- python dictionary containing your updated parameters
                  parameters["W" + str(l)] = ...
                  parameters["b" + str(l)] = ...
    """

    L = len(parameters) // 2 # number of layers in the neural network

    # Update rule for each parameter. Use a for loop.
    ### START CODE HERE ### (≈ 3 lines of code)
    for l in range(L):
        parameters["W" + str(l+1)] = parameters["W" + str(l+1)]-learning_rate*grads["dW" + str(l+1)]
        parameters["b" + str(l+1)] = parameters["b" + str(l+1)]-learning_rate*grads["db" + str(l+1)]
    ### END CODE HERE ###

    return parameters
```

結果：

```
W1 = [[-0.59562069 -0.09991781 -2.14584584  1.82662008]
 [-1.76569676 -0.80627147  0.51115557 -1.18258802]
 [-1.0535704  -0.86128581  0.68284052  2.20374577]]
b1 = [[-0.04659241]
 [-1.28888275]
 [ 0.53405496]]
W2 = [[-0.55569196  0.0354055   1.32964895]]
b2 = [[-0.84610769]]
```